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Multi-layer security based multiple image encryption technique☆

Shazia Sabir, Vandana Guleria *

Department of Mathematics, Birla Institute of Technology Mesra, Ranchi, India

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ABSTRACT

This paper designs a novel multiple image encryption technique utilizing affine Hill cipher (AHC), reality-preserving two-dimensional discrete fractional Hartley transform (RP2DFrHT) and generalized two-dimensional (2D) Arnold map (AM). In our technique, three indexed images are obtained from three color images. Afterwards, AHC performs a strong degree of confusion and diffusion operations and RP2DFrHT provides real domain output image. Lastly, 2D AM dislocates the image pixel position. After the encryption process, single-channel real-valued encrypted image is obtained which gives convenience in the display, storage and transmission of an image over an unsecured network. The proposed technique is multi-layer secured in the time, frequency and co-ordinate domains. Moreover, the secret keys, arrangements and positions of used parameters all are imperative for the correct decryption process of our technique. Simulation results, security analysis, statistical analysis and comparison analysis ensure that the proposed technique has an excellent encryption effect, sufficient security and robustness.

1. Introduction

We live in a digital world where every day, our lives have been prolifically impacted by the rapid development of new technologies. In 5G/6G networks, an unprecedentedly large amount of digital images are being processed, stored and transmitted over an unsecured channel. The massive inflow of digital images in terms of velocity, volume and veracity is a threat to security, efficiency and capacity space. Therefore, the security of image data during storage and transmission is a major concern. In this context, encryption is one of the best methods of preserving privacy and a secure communication system. Image encryption techniques can be classified into two kinds of encryption. One is chaos-based encryption and the other is transform-based encryption. Owing to the properties of ergodicity, transitivity, initial conditions, non-periodicity, simplicity and pseudo-randomness, chaos-based encryption has been effective in the field of image encryption. As a result, image encryption techniques [1,2] based on permutation and diffusion operations have been developed. However, these techniques are generally not enough to provide a high level of security because their chaotic dynamic properties degrade rapidly with finite precision in digital computers. To enhance the level of security and boost the dynamic degradation, Xian and Wang have contributed a remarkable work using the concept of fractal sorting matrix [3,4] and fractal sorting vector with spatiotemporal chaotic system [5]. Recently, numerous studies have been conducted based on transform-based encryption, such as fractional Fourier transform [6], fractional Hartley transform [7], etc. Apart from their usefulness and wide applicability, yet, most of the fractional transform-based techniques produce complex output, which makes the storage, display and transmission of digital information more difficult. To address these limitations, Venturini and Duhamel [8] developed a reality-preserving fractional transform. The reality-preserving property provides the real domain output image corresponding to the real domain input image and makes the system effective for storing, displaying and transmitting of

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* Corresponding author.
E-mail address: vandana@bitmesra.ac.in (V. Guleria).

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data in the digital environment. However, the aforementioned techniques only process one image at a time which greatly reduces the encryption efficiency. To conquer the disadvantage of single-image encryption, multiple-image encryption (MIE) has grabbed a great deal of attention. Researchers have proposed several MIEs [9,10]. The existing MIEs improve the encryption efficiency but are merely applicable for one type of image data, such as grayscale image or RGB image. Also, these techniques increase the size of the encrypted images which reduces the encryption capacity and increases the transmission cost.

Our contributions: To address the limitations of the existing techniques, our proposed technique integrates the three fascinating algorithms such as AHC, RP2DFrHT and 2D AM into an effective real-valued, single-channel cryptographic mechanism. The novelty and contributions of our paper are highlighted in the subsequent paragraphs.

1. The existing chaos-based techniques [1,2] perform the permutation and diffusion operations in the same domain, which significantly reduces the security level of the system. To overcome this limitation, we have used transform-based encryption in our proposed technique. This approach transforms the original image data from the spatial domain to different fractional domains, which improves the security of the system and makes our system more effective.
2. The existing fractional transform-based techniques [6,7] convert the whole input data from the real field to complex field. Therefore, the storage and transmission increase complexity and overheads in a digital channel. To overcome this limitation, the second level of encryption uses RP2DFrHT. The reality-preserving approach gives the guarantee of real-valued output corresponding to real-valued input and improves the encryption efficiency of the proposed technique. Therefore, our proposed technique is efficient in terms of time complexity, storage complexity and communication complexity.
3. The existing Hill cipher-based techniques [11,12] select Hill cipher (HC) keys from involutory and self-invertible matrices. Besides that, in these techniques, the large size of the image needs a large size of the secret key which is not conductive for the execution of encryption and decryption processes. Therefore, these techniques are not only vulnerable to brute force attacks but also not enough to provide a high level of security. However, the proposed technique selects HC keys from $SL_m(F_p)$ and $M_m(F_p)$ domains. Therefore, our technique facilitates exorbitant huge key space which is large enough to resist brute force attacks and provides a high level of security.
4. In existing techniques [13,14], the periodic property of AM is exploited for the decryption process. These techniques are very time-consuming. In order to overcome the above limitation, we have utilized the inverse AM for the decryption procedure of our proposed technique. This makes the system faster and provides a high level of security and randomness. In addition, it also enlarges the key space.
5. The existing techniques [15,16] provide one layer of security because the decryption procedures of these techniques depend upon secret keys only. If an unauthorized person has access to secret keys, the encryption technique is no more secure. However, our proposed technique is three-layer secured. The first layer of security is attained by the Hill cipher keys, the second layer of security is attained by the arrangements of AHC parameters and the third layer of security is attained by the positions of AHC parameters. If an unauthorized person has access to secret keys still he is unable to recover the original image from the encrypted image without knowing the correct arrangements and positions of the AHC parameters.
6. The existing techniques [17,18] can encrypt single or double images and are merely applicable either for the grayscale image or RGB image. However, our proposed technique encrypts multiple images simultaneously and is applicable for three types of image data such as RGB image, grayscale image and binary image.
7. The existing techniques [9,19] increase the dimension of the encrypted data which reduces the encryption capacity and increases the transmission cost. However, the dimension of the encrypted data of the proposed technique is equal to the original data.

Paper Organization: The remaining sections are assembled as follows. Section 2 discusses the mathematical formulations of AHC, RP2DFrHT and generalized 2D AM. The detailed discussion of the presented technique is illustrated in Section 3. Section 4 conducts the simulation analysis. Sections 5 and 6 examine the security and statistical analyses, respectively. Section 7 illustrates the comparison analysis with existing similar techniques. The paper is concluded in the last section.

2. Preliminaries

In this section, the mathematical formulations of AHC, RP2DFrHT and 2D AM are provided.

2.1. AHC [20]

AHC is one of the well-known polygraphic ciphers which extends the concept of Hill cipher by combining it with non-linear affine transformation. In our proposed technique, we have used AHC to secure multiple RGB images. Consider an RGB image of size $n \times n \times 3$ which is segregated into three RGB planes. The segregated planes are further decomposed into $m \times m$ block matrix such that the order of the block matrix divides the order of each plane matrix, i.e., $m|n$. If the block matrix is chosen in such a way that the order of the block matrix does not divide the order of the plane matrix, i.e., $m \nmid n$, then, in this case, some redundant rows or columns or both are added to the original image matrix. During the implementation of AHC, two types of keys are used. One is the multiplicative key which is chosen from a special linear domain over a finite field $\mathbb{F}_p$, i.e., $SL_m(\mathbb{F}_p)$ and another is the additive key which is chosen from $M_m(\mathbb{F}_p)$ domain. The size of each AHC key and block matrix should be equal. Here, the domain $SL_m(\mathbb{F}_p)$

defines as the subgroup of general linear group $GL_m(\mathbb{F}_p)$ and the domain $M_m(\mathbb{F}_p)$ represents the set of all $m \times m$ matrices over finite field $\mathbb{F}_p$. The mathematical formulations of $GL_m(\mathbb{F}_p)$, $SL_m(\mathbb{F}_p)$ and $M_m(\mathbb{F}_p)$ are given below.

$$GL_m(\mathbb{F}_p) = \{L = [l_{i,j}]_{m\times m} \mid l_{i,j} \in \mathbb{F}_p, \quad \det(L) \neq 0\}. \tag{1}$$

$$SL_m(\mathbb{F}_p) = \{L \in GL_m(\mathbb{F}_p) \mid \det(L) = 1\}. \tag{2}$$

$$M_m(\mathbb{F}_p) = \{L = [l_{i,j}]_{m\times m} \mid l_{i,j} \in \mathbb{F}_p\}. \tag{3}$$

The mathematical expression of AHC for $m \times m$ block matrix is given below.

$$E_{m\times m} = B_{m\times m}.M_{m\times m} + A_{m\times m} \pmod{256}, \tag{4}$$

where $B_{m\times m}$ denotes the block matrix of image data, $M_{m\times m}$ is the multiplicative key chosen from $SL_m(\mathbb{F}_p)$ domain, $A_{m\times m}$ is the additive key chosen from $M_m(\mathbb{F}_p)$ domain and $E_{m\times m}$ is the encrypted block matrix of size $m \times m$. The mathematical expression of inverse AHC (iAHC) for $m \times m$ encrypted block matrix is given below.

$$B_{m\times m} = (E_{m\times m} - A_{m\times m}).M^{-1}_{m\times m} \pmod{256}, \tag{5}$$

where $M^{-1}_{m\times m}$ is the multiplicative inverse of $M \in SL_m(F_p)$.

2.2. RP2DFrHT [21]

The mathematical model of discrete Hartley transform (DHT) matrix H of size $n \times n$ is expressed as follows.

$$H_{ab} = \frac{1}{\sqrt{n}}\left[\cos\left(\frac{2\pi ab}{n}\right) + \sin\left(\frac{2\pi ab}{n}\right)\right], \tag{6}$$

where $a, b = 0, 1, \dots.., n-1$.

Tridiagonal matrix S is defined as follows.

$$S = \begin{bmatrix} 2 & 1 & 0 & \dots & 0 & 1 \\ 1 & 2\cos\left(\frac{2\pi}{n}\right) & 1 & \dots & 0 & 0 \\ 0 & 1 & 2\cos\left(\frac{4\pi}{n}\right) & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2\cos\left[\frac{2(n-2)\pi}{n}\right] & 1 \\ 1 & 0 & 0 & \dots & 1 & 2\cos\left[\frac{2(n-1)\pi}{n}\right] \end{bmatrix}. \tag{7}$$

The eigenvalue decomposition of the matrix H is defined as follows.

$$H = UDU^t, \tag{8}$$

where the matrix U is the eigenvector of matrix S, D denotes the diagonal matrix whose diagonal entries consist of the eigenvalues of the matrix H and U^t signifies the transposed matrix of U.

The cth order of DFrHT matrix is defined as follows.

$$H^c = UD^cU^t, \tag{9}$$

where the parameter c denotes the fractional order.

To encrypt the $N \times N$ matrix P, 2DFrHT is defined as follows.

$$K = H^c P (H^c)^t. \tag{10}$$

The 2DFrHT results in complex domain output corresponding to real domain input which is inconvenient for storage, display and transmission.

To obtain the real domain output corresponding to real domain input, RP2DFrHT is used. The kernel matrix of RP2DFrHT is illustrated below.

$$R^c_H = \begin{bmatrix} \mathrm{Re}(H^c) & -\mathrm{Im}(H^c) \\ \mathrm{Im}(H^c) & \mathrm{Re}(H^c) \end{bmatrix}. \tag{11}$$

To encrypt the $N \times N$ matrix P, RP2DFrHT is implemented as follows.

$$W = R^c_H P R^d_H, \tag{12}$$

where the fractional orders are represented by c and d.

To decrypt the $N \times N$ matrix W, IRP2DFrHT is implemented as follows.

$$P = R^{-c}_H W R^{-d}_H. \tag{13}$$

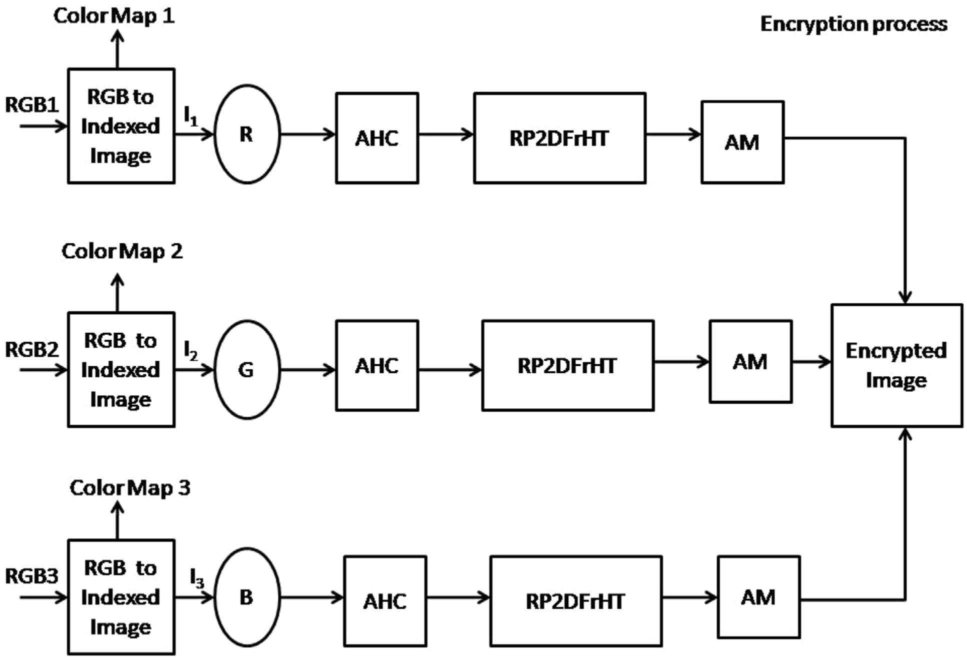


Fig. 1. Block diagram of the encryption process.

2.3. Generalized 2D AM [22]

The generalized AM is an excessively utilized scrambling algorithm in image encryption. The mathematical expression of AM is demonstrated below.

$$\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 & \beta \\ \alpha & \alpha\beta + 1 \end{bmatrix} \begin{bmatrix} o_1 \\ o_2 \end{bmatrix} \pmod{N}, \tag{14}$$

The reverse procedure of AM is demonstrated in the following equation.

$$\begin{bmatrix} o_1 \\ o_2 \end{bmatrix} = \begin{bmatrix} \alpha\beta + 1 & -\beta \\ -\alpha & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \pmod{N}, \tag{15}$$

Here, Arnold keys are represented as α and β, the co-ordinates (o_1, o_2) and (e_1, e_2) are pixel positions before and after dislocation and α, $\beta \in \mathbb{Z}^+$, $\mathbb{Z}^+$ is the set of positive integer.

3. Explanation of the proposed technique

Based on AHC, RP2DFrHT and 2D AM, a novel image encryption technique is designed to secure multiple color, grayscale and binary images. In this section, we have discussed the detailed procedures of the proposed encryption and decryption technique.

3.1. Encryption process of RGB image

Firstly, the encryption process of the color images is discussed. Fig. 1 illustrates the block diagram of the encryption process. The detailed discussion of each stage of the encryption process is discussed below.

Stage-I: Transformation of RGB images into indexed images

Initially, three color images are transformed into three indexed images as I_1, I_2 and I_3. The indexed images are considered as red (R), green (G) and blue (B) planes of the single RGB image.

Stage-II: Image blocking (preprocessing of AHC)

Before applying AHC, each plane of the single RGB image is divided into small non-overlapping blocks with different block sizes.

Stage-III: Perform AHC on each block matrix

In this stage, following the steps outlined in Section 2.1, AHC is applied on each non-overlapping block. During the implementation of AHC, the multiplicative and additive keys are used as secret keys.

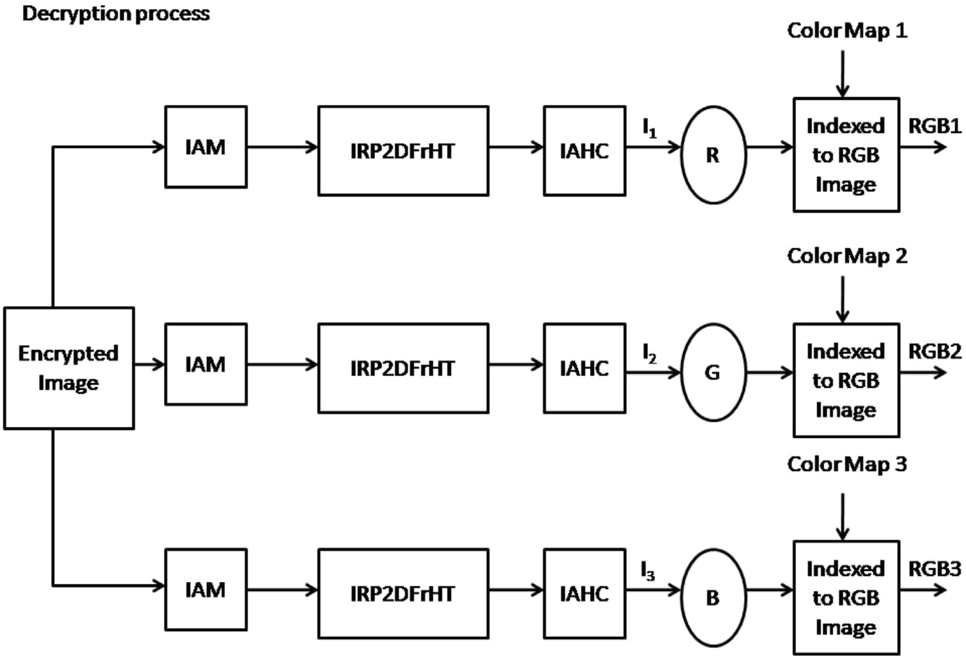


Fig. 2. Block diagram of the decryption process.

Stage-IV: RP2DFrHT

After AHC, following the steps outlined in Section 2.2, RP2DFrHT is applied on the output images of the third stage. The fractional orders are used as secret keys in this stage.

Stage-V: 2D AM

In the last stage, following the steps outlined in Section 2.3, 2D AM is applied to dislocate the positions of former image pixels. The Arnold keys are treated as secret keys in this stage.

The output images of the last stage are then merged to obtain the final encrypted image. Therefore, the single color encrypted image is obtained corresponding to three color images.

3.2. Decryption process of RGB image

To attain the original color images, the encryption process has been reversed. Fig. 2 illustrates the block diagram of the decryption process.

Stage-I: Inverse AM

Initially, the final encrypted image is decomposed into three RGB planes. Following the steps outlined in Section 2.3, inverse AM is applied on each plane of the final encrypted image. During implementation, same encryption keys are used.

Stage-II: Inverse RP2DFrHT

In the second stage of decryption, following the steps outlined in Section 2.2, IRP2DFrHT is applied on each plane of the partially decrypted image. During implementation, the inverse fractional orders are used.

Stage-III: Image blocking (preprocessing of inverse AHC)

In this stage, the second stage output images are partitioned into non-overlapping blocks whose size remains the same as the encryption process.

Stage-IV: Inverse AHC

In the last stage of decryption, inverse AHC is implemented on each plane of the intermediate decrypted image by following the procedure stated in Section 2.1. During implementation, the inverse multiplicative key and inverse additive key are utilized as decryption keys.

The output images of the last stage are then merged to their color maps to get three original images.

3.3. Encryption and decryption processes of grayscale and binary images

This section discusses the encryption and decryption processes of grayscale and binary images.

3.3.1. Preprocessing of grayscale and binary images

Figs. 1 and 2 demonstrate the block diagram of the proposed technique which is only for the RGB images. However, the proposed technique is developed to secure grayscale and binary images as well. Therefore, we perform preprocessing on the grayscale and binary images before applying the given block diagram. The procedure of preprocessing is illustrated below.

Transformation of grayscale images into RGB images

As we know, each pixel of an RGB image contains 24 bits, whereas each pixel of a grayscale image contains 8 bits. In this way, three-bit planes of the grayscale images give a single RGB image. Since the proposed technique encrypts three color images at once, therefore we have used another six grayscale images to obtain two RGB images. The resulting color images are encrypted together by following the same procedure stated in Section 3.1. Therefore, we get the single color encrypted image corresponding to nine grayscale images. To get back the grayscale images, the resulting encrypted image is decrypted by following the same procedure stated in Section 3.2.

Transformation of binary images into RGB images

For the conversion of binary images into RGB images, we first convert the binary images into the grayscale images and then the resulting grayscale images are further converted into the color images. As we know, each pixel of a binary image contains one bit. Therefore, eight-bit planes of binary images give a single grayscale image. Following this concept, we have used another sixteen binary images to get two grayscale images. The resulting three grayscale images are then combined to obtain the single color image. In this way, twenty-four binary images give the single color image. Similarly, to obtain two color images, we have used another forty-eight binary images. The resulting color images are encrypted together by following the same procedure stated in Section 3.1. Hence, we get the single color encrypted image corresponding to seventy-two binary images. To get back the binary images, the resulting encrypted image is decrypted by following the same procedure stated in Section 3.2.

4. Experimental results

This section demonstrates the experimental results to testify the effectiveness and validity of our proposed technique. The simulation is done on three types of images. The first experiment is performed on three RGB images, the second experiment is performed on nine grayscale images and the third experiment is performed on seventy-two binary images. The detailed discussion of the whole process is provided below.

4.1. Results of RGB images

To show the encryption results of the RGB images, we have chosen three standard RGB images, Baboon, Lena and Peppers, as displayed in Fig. 3(a–c), respectively. The size of each image is $512 \times 512 \times 3$. Initially, three RGB images are converted to three indexed images by extracting their color maps. The indexed images corresponding to three RGB images are treated as R, G and B planes of the single RGB image. After conversion, AHC is implemented on each plane of an RGB image using multiplicative and additive keys. The multiplicative keys for R, G and B planes are given below.

$$M_R = \begin{bmatrix} 13 & 16 & 128 & 80 \\ 4 & 5 & 160 & 139 \\ 0 & 0 & 20 & 3 \\ 0 & 0 & 53 & 8 \end{bmatrix}, \; M_G = \begin{bmatrix} 2 & 7 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 4 & 2 & 2 & 0 & 3 & 3 & 4 \\ 0 & 0 & 5 & 2 & 7 & 6 & 8 & 9 \\ 0 & 0 & 7 & 3 & 4 & 5 & 3 & 9 \\ 0 & 0 & 0 & 0 & 1 & 5 & 7 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \; M_B = \begin{bmatrix} 25 & 31 & 60 & 39 \\ 4 & 5 & 151 & 91 \\ 0 & 0 & 12 & 31 \\ 0 & 0 & 5 & 13 \end{bmatrix}.$$

The additive keys for R, G and B planes are given below.

$$A_R = \begin{bmatrix} 113 & 118 & 150 & 81 \\ 101 & 119 & 88 & 218 \\ 104 & 190 & 189 & 204 \\ 71 & 185 & 124 & 128 \end{bmatrix}, \; A_G = \begin{bmatrix} 145 & 52 & 210 & 140 & 19 & 55 & 158 & 22 \\ 136 & 39 & 228 & 165 & 61 & 106 & 92 & 61 \\ 48 & 53 & 102 & 51 & 186 & 224 & 85 & 78 \\ 70 & 101 & 25 & 27 & 6 & 127 & 230 & 158 \\ 109 & 72 & 60 & 69 & 216 & 121 & 8 & 31 \\ 53 & 215 & 95 & 74 & 170 & 53 & 206 & 168 \\ 196 & 100 & 138 & 98 & 113 & 113 & 212 & 24 \\ 45 & 43 & 61 & 118 & 134 & 145 & 185 & 152 \end{bmatrix}, \; A_B = \begin{bmatrix} 115 & 207 & 7 & 210 \\ 181 & 77 & 173 & 142 \\ 166 & 162 & 116 & 143 \\ 210 & 46 & 111 & 200 \end{bmatrix}.$$

In the second stage, RP2DFrHT is implemented on each partially encrypted image using fractional orders. The fractional orders for R, G and B planes are $[c_R, d_R]$=[0.7683, 0.4521], $[c_G, d_G]$=[0.3952, 0.6745] and $[c_B, d_B]$=[0.8554, 0.5987], respectively. In the last stage, AM is implemented on each intermediate encrypted image using Arnold keys. The Arnold keys for R, G and B planes

Fig. 3. Encryption and decryption results of RGB images.

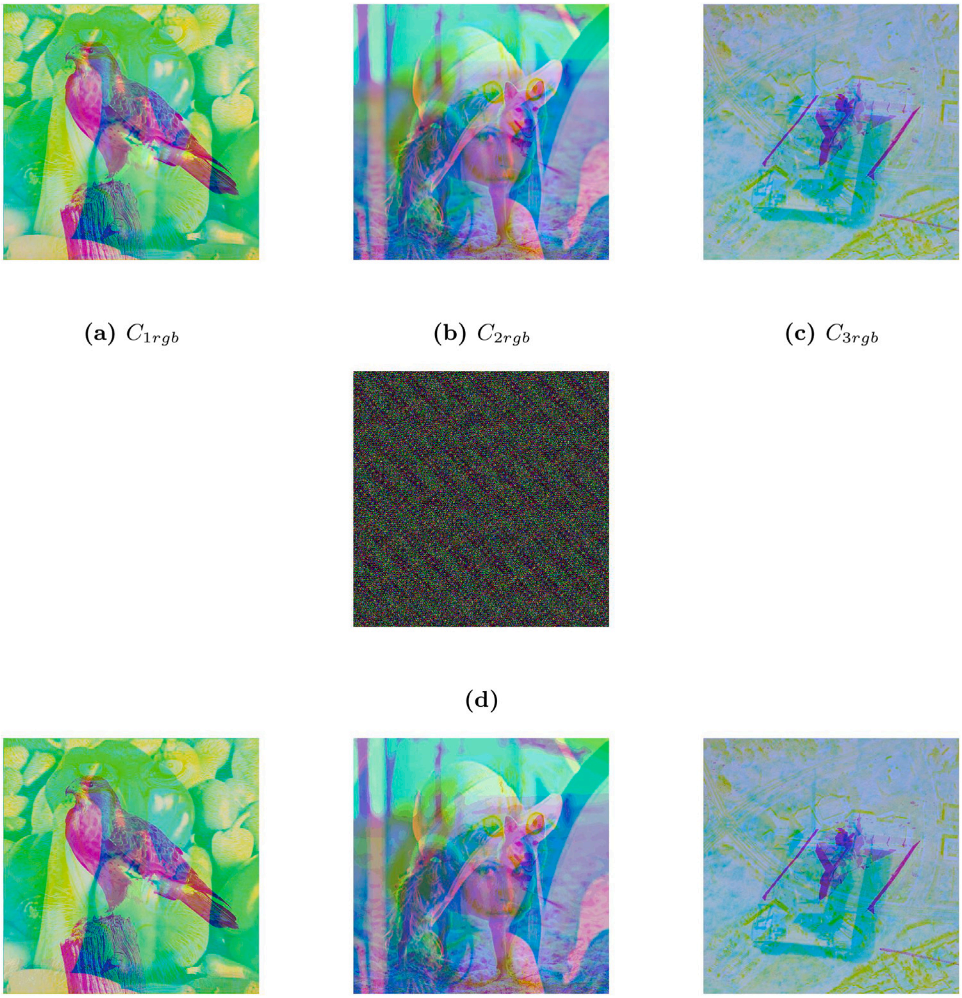
are $[\alpha_R, \beta_R]$=[126, 97], $[\alpha_G, \beta_G]$=[65, 170] and $[\alpha_B, \beta_B]$=[151, 83], respectively. The proposed encryption procedure provides the single encrypted image corresponding to three RGB images, as shown in Fig. 3(d). In the decryption process, firstly, inverse AM is implemented to get the partially decrypted image. Subsequently, IRP2DFrHT gives the intermediate decrypted image and lastly, inverse AHC gives the final decrypted image. Fig. 3(e–g) display the final decrypted images after the last stage of decryption.

4.2. Results of grayscale images

To show the encryption results of the grayscale images, we have chosen nine grayscale images ($G_1 - G_9$). As we know, three grayscale images are concatenated to get a single RGB image. In this way, the grayscale images ($G_1 - G_3$) gives the RGB image C_{1rgb}, the grayscale images ($G_4 - G_6$) gives the RGB image C_{2rgb} and the grayscale images ($G_7 - G_9$) gives the RGB image C_{3rgb}. Therefore, we get three RGB images (C_{1rgb}–C_{3rgb}). The resulting RGB images, as shown in Fig. 4(a–c), are then encrypted together by following the same procedure stated in Section 4.1. The encryption procedure provides the single encrypted image corresponding to nine grayscale images, as illustrated in Fig. 4(d). In the decryption, the same procedure stated in Section 4.1 is followed. Fig. 4(e-g) display the final decrypted images.

4.3. Results of binary images

To show the encryption results of the binary images, we have chosen 72 binary images ($B_1 - B_{72}$). Keeping in view of length of the paper size, we have provided only the final results of the binary images. As we know, eight binary images are concatenated to



(a) C_{1rgb} (b) C_{2rgb} (c) C_{3rgb}

(d)

(e) (f) (g)

Fig. 4. Results of grayscale images.

get a single grayscale image. Therefore, the binary images ($B_1 - B_8$) gives the grayscale image G_{1B}. In this way, we get the nine grayscale images ($G_{1B} - G_{9B}$) corresponding to the 72 binary images ($B_1 - B_{72}$). The resulting grayscale images are then concatenated to obtain the color image. The resulting color images ($C_{1G} - C_{3G}$), as shown in Fig. 5(a–c), are then encrypted together by following the same procedure stated in Section 4.1. After implementation, the single encrypted image, as illustrated in Fig. 5(d), is obtained corresponding to 72 binary images. To get back the binary images, the resulting encrypted image is decrypted by following the same procedure stated in Section 4.1. Fig. 5(e–g) display the final decrypted images.

5. Security analysis

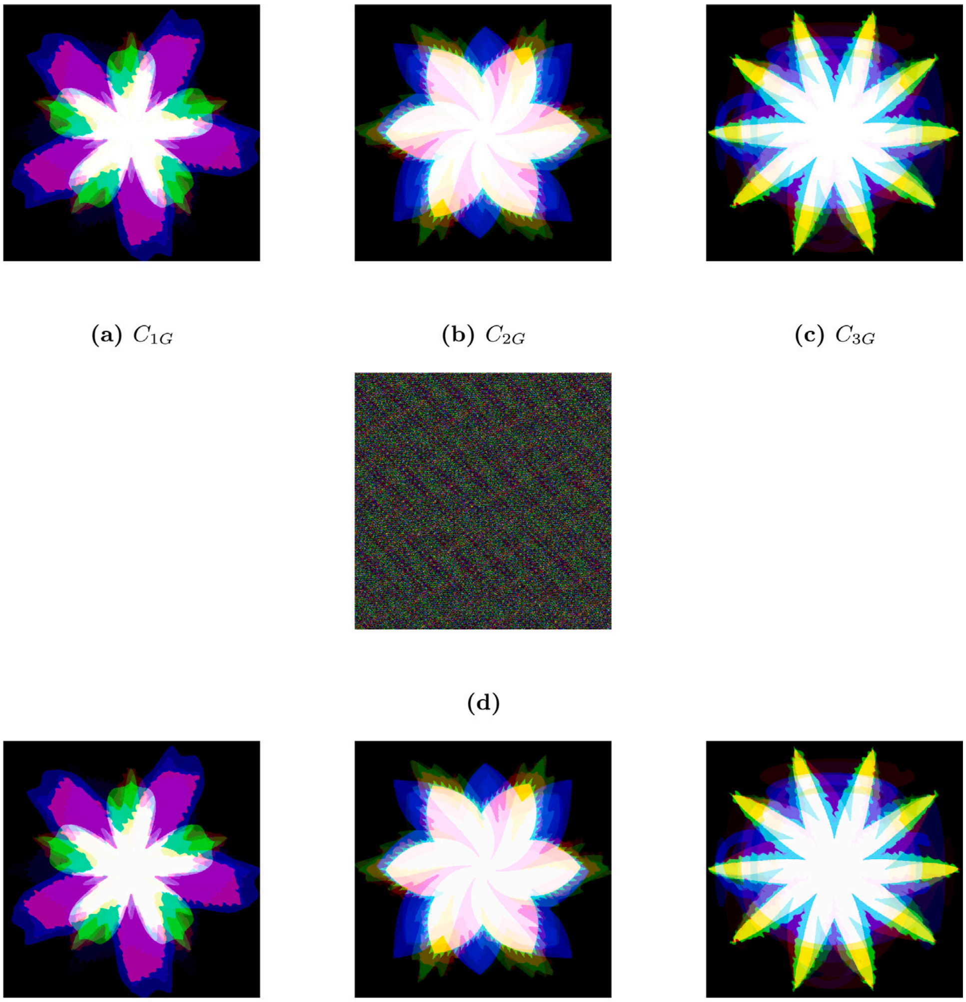
5.1. Key sensitivity

Sensitivity analysis is used to measure the resistance of encryption scheme to brute force attacks. The analysis is carried out in two phases: (i) the key sensitivity of encryption key and (ii) the key sensitivity of decryption key.

5.1.1. Key sensitivity of encryption key

To examine the key sensitivity of encryption key, the following steps are adopted.

1. Initially, the original image and correct secret keys are taken.



(a) C_{1G} (b) C_{2G} (c) C_{3G}

(d)

(e) (f) (g)

Fig. 5. Encryption and decryption results of binary images.

2. The correct secret keys are slightly changed to obtain the modified secret keys.
3. Following the proposed technique, the original image is encrypted using the correct secret keys and modified secret keys.
4. Two encrypted images are obtained corresponding to correct and modified secret keys.

In our experiment, following the above procedure, correct Arnold keys and fractional orders are slightly changed to obtain their modified secret keys. Fig. 6(a) and (b) show the encrypted images corresponding to correct and modified Arnold keys, respectively. Fig. 6(c) reflects the difference between the resulting encrypted images. Similarly, Fig. 6(d) and (e) show the encrypted images corresponding to correct and modified fractional orders, respectively. Fig. 6(f) reflects the difference between the resulting encrypted images. The test results show that the encryption results are completely changed by modifying the encryption keys. To estimate the difference between the resulting encrypted images, NPCR (number of pixel change rate) is used. Table 1 shows the NPCR values which validate that our technique is highly sensitive with respect to encryption keys.

5.1.2. Key sensitivity of decryption keys

In this paper, we have discussed two sensitivity analyses with respect to decryption keys. (i) In sensitivity analysis-I, the wrong fractional orders are taken while the remaining secret keys are kept unchanged. Fig. 7(a–c) display the test results of sensitivity analysis-I. (ii) In sensitivity analysis-II, the incorrect positions of multiplicative keys are taken, whereas the remaining parameters are kept constant. Fig. 7(d–f) display the test results of sensitivity analysis-II. In order to further illustrate the test results, correlation,

Table 1
NPCR values between two encrypted images.

Secret keys	Parameter	R	G	B
Arnold keys	NPCR	99.9996	99.9996	99.9969
Fractional orders	NPCR	100	100	100

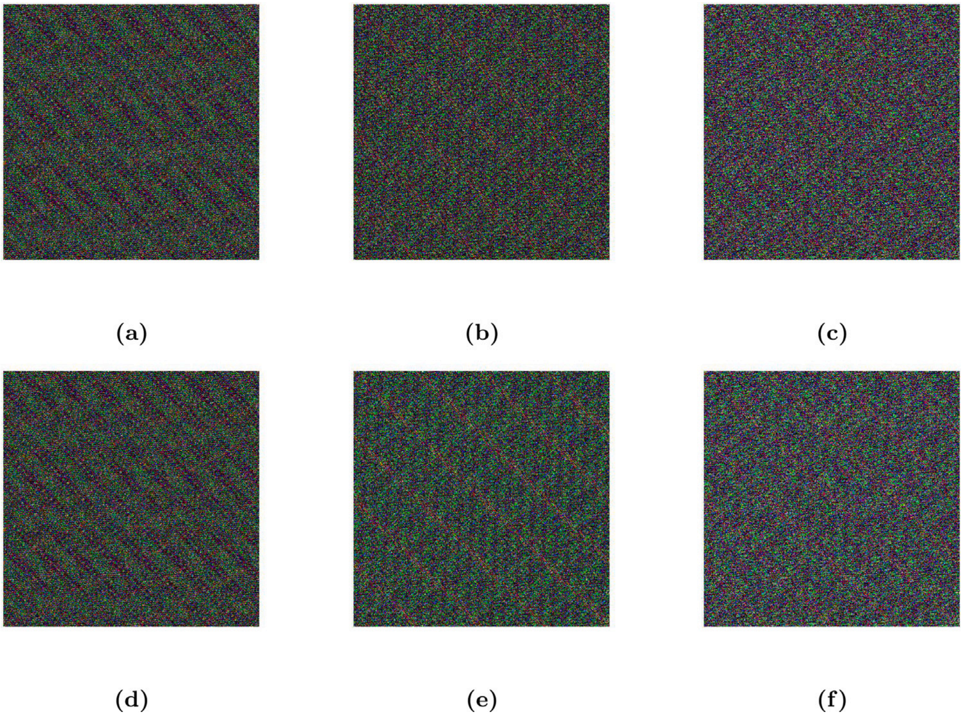


Fig. 6. Test results of sensitivity analysis of encryption keys: (a) encrypted image with correct Arnold keys (b) encrypted image with modified Arnold keys (c) difference of Figs. 6(a) and 6(b) (d) encrypted image with correct fractional orders (e) encrypted image with modified fractional orders (f) difference of Figs. 6(d) and 6(e).

MSE and PSNR are computed as shown in Table 5. Therefore, the test results validate that our technique is extremely sensitive with respect to decryption keys.

5.2. Noise attack

In real-time processing over an open network, some information may be corrupted by noise attacks. To test the resistance of the proposed technique to noise attack, Gaussian noise is added to the red plane of the encrypted image (Fig. 3(d)) with mean 0 and variance 0.01, while the other two planes are kept noise free. The test results of Gaussian noise are displayed in Fig. 8(a–c). Therefore, after analyzing the test results, it is observed that the proposed technique shows good performance against noise attacks.

5.3. Occlusion attack

In real-time processing over an open network, some information may be attacked and lost. To check the feasibility of our proposed technique against occlusion attacks, we have occluded 5% of the data of the encrypted image (Fig. 3(d)) from the upper left corner. The corresponding decrypted images are displayed in Fig. 9(a–c). The test results validate that the resulting decrypted images are still visible to the naked eye. Therefore, the proposed technique is strong enough to resist occlusion attacks.

6. Statistical analysis

In order to analyze the robustness of the proposed technique, statistical analysis has been carried out based on statistical parameters such as histogram analysis, MSE, PSNR, correlation coefficient, SSIM, entropy, key space and classical attacks.

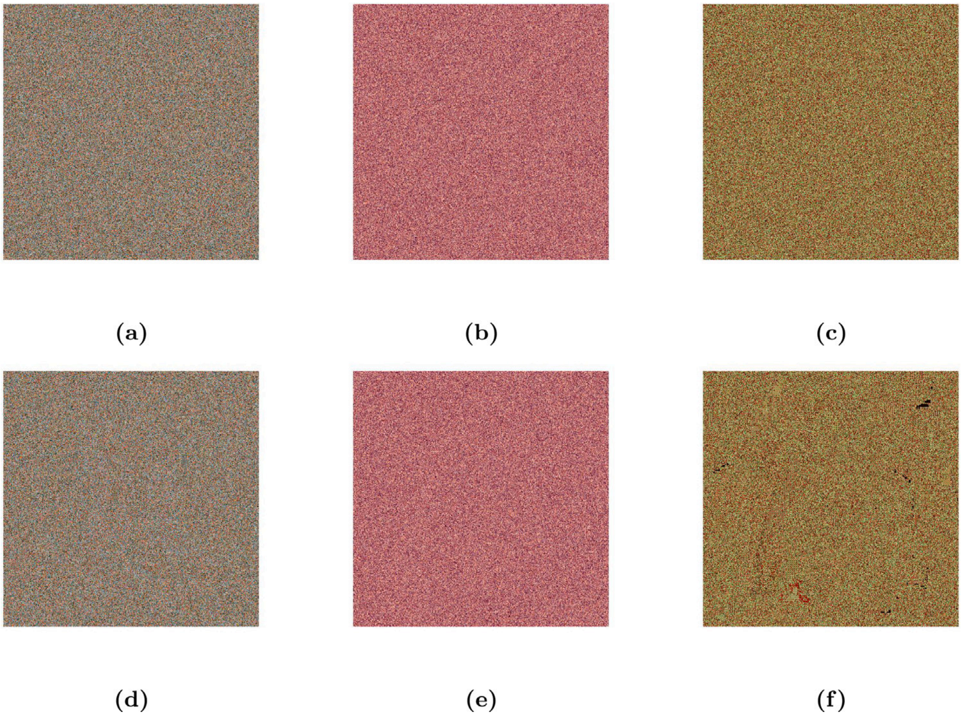


Fig. 7. Decryption results: (a–c) Results of sensitivity analysis-I (d–f) Results of sensitivity analysis-II.

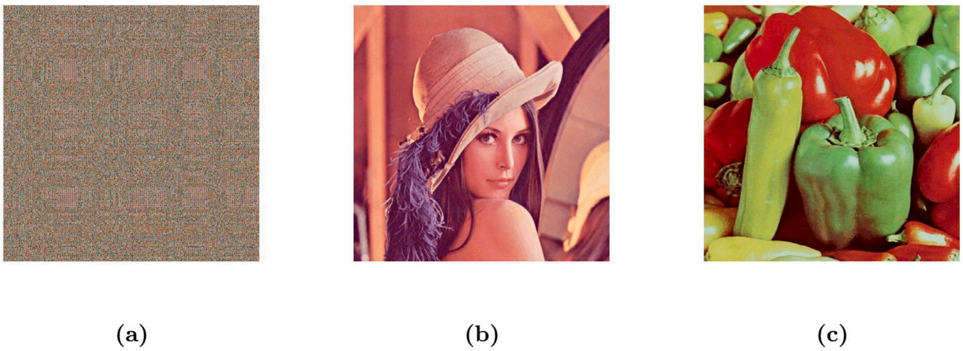


Fig. 8. Results of noise attack: (a–c) Results of Gaussian noise.

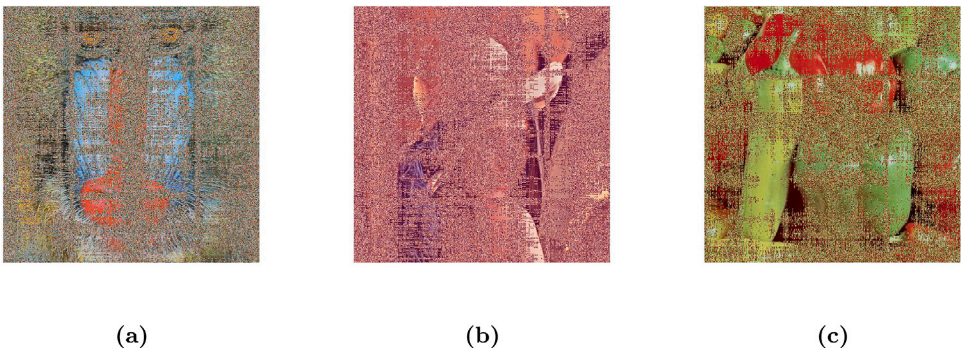


Fig. 9. Results of occlusion attack: (a–c) The decrypted images corresponding to three color images, when upper left corner of the encrypted image is occluded 5%.

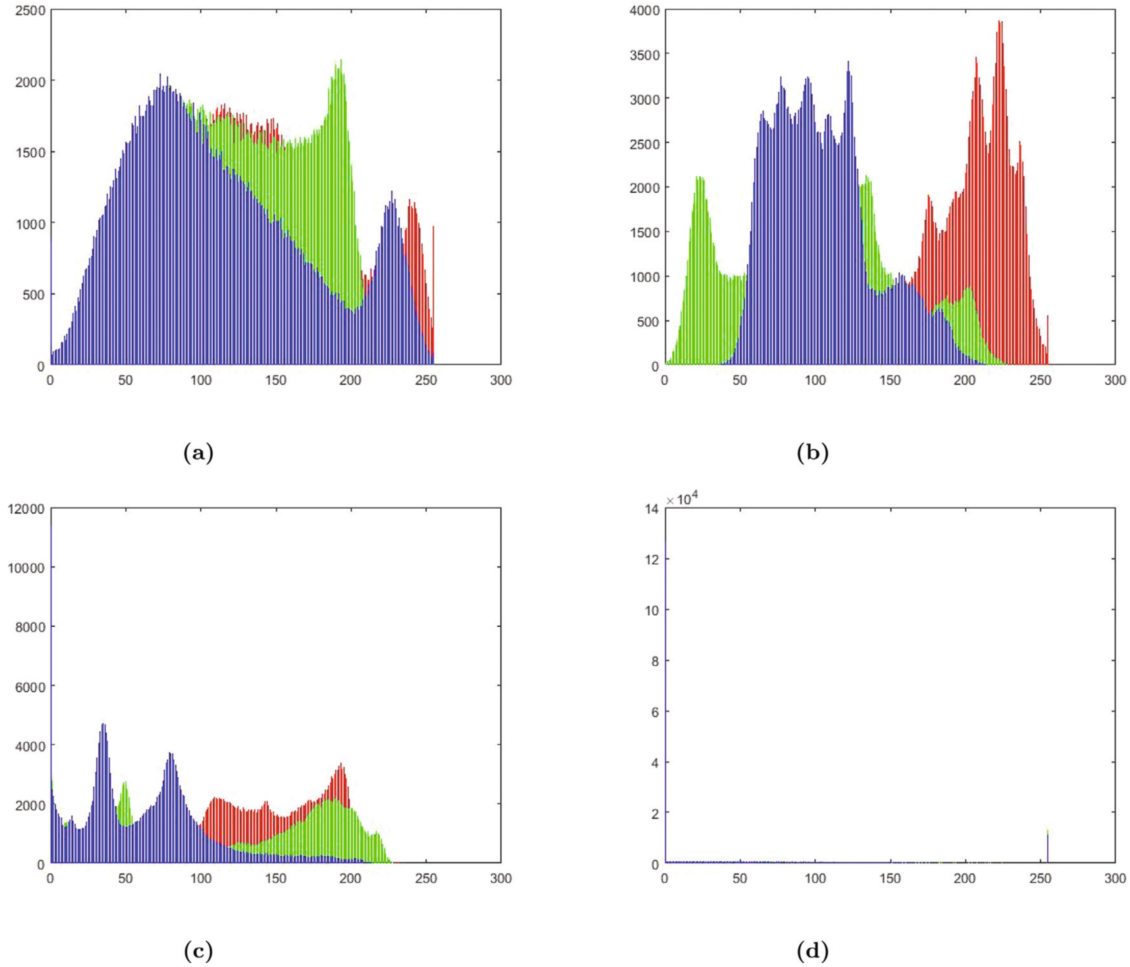


Fig. 10. Histogram analysis of RGB images: (a) histogram of Fig. 3(a) (b) histogram of Fig. 3(b) (c) histogram of Fig. 3(c) (d) histogram of Fig. 3(d).

6.1. Histogram analysis via graphical method

Image histogram is a graphical representation of pixel intensity distribution. Fig. 10(a–c) display the histograms of the Baboon, Lena and Peppers color images. Fig. 10(d) displays the histogram of the cipher image corresponding to 3 color images. Similarly, Fig. 11(a) displays the histogram of the encrypted image (Fig. 4(d)) corresponding to 9 grayscale images and Fig. 11(b) displays the histogram of the encrypted image (Fig. 5(d)) corresponding to 72 binary images. By observing all the histograms of the encrypted images corresponding to different types of images, it is clear that the distribution of pixels in the encrypted image is entirely different from the distribution of pixels in the original image. Therefore, the proposed technique is robust against histogram-based attacks.

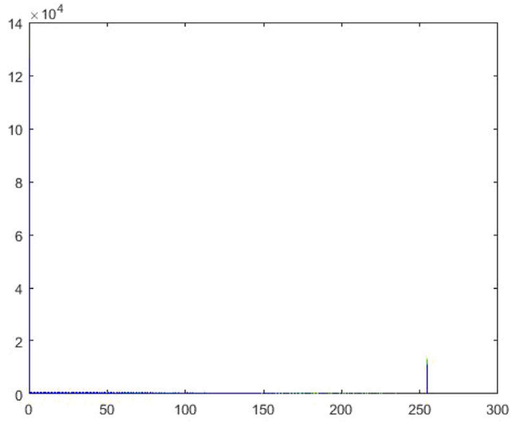
6.2. Correlation coefficient, mean squared error (MSE), peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM)

The mathematical formulation of the correlation coefficient is computed using Eq. (16).

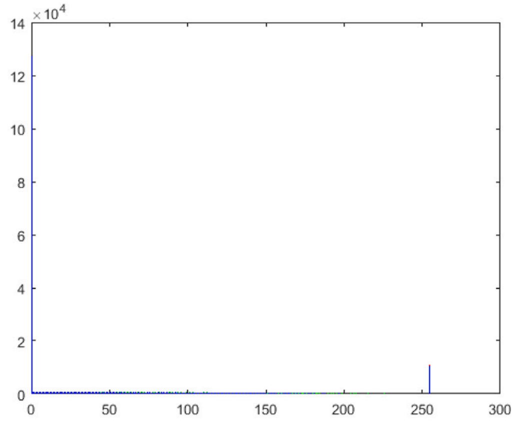
$$C = \frac{\sum_{s=1}^{S}\sum_{t=1}^{T}(X(s,t)-\bar{X})(Y(s,t)-\bar{Y})}{\sqrt{\left(\sum_{s=1}^{S}\sum_{t=1}^{T}(X(s,t)-\bar{X})^2\right)\left(\sum_{s=1}^{S}\sum_{t=1}^{T}(Y(s,t)-\bar{Y})^2\right)}}, \tag{16}$$

where X and Y are the input and output images, respectively and $\bar{X}$ and $\bar{Y}$ are the mean of X and Y, respectively.

The value of C must be in the range of -1 and 1 i.e., $-1 \leq C \leq 1$. When $C = 1$, the input and output images have a strong linear relationship. The input and output images are negatively linked, when $C = -1$. When $C = 0$, no relationship exists between the input and output images.



(a)



(b)

Fig. 11. Histogram analysis of grayscale and binary images: (a) histogram of Fig. 4(d) (b) histogram of Fig. 5(d).

Table 2
Correlation coefficient, MSE and PSNR of encrypted and decrypted images of RGB image.

Parameters	Encrypted image			Decrypted image		
	R	G	B	R	G	B
Correlation	−0.0015	−0.0051	5.3176e−04	1	1	1
MSE	3.9285e+04	3.8766e+04	4.1447e+04	0	0	0
PSNR	2.1886	2.2463	1.9559	Inf	Inf	Inf

The mathematical formulation of MSE is computed using Eq. (17).

$$\mathrm{MSE} = \frac{1}{S \times T} \sum_{s=1}^{S} \sum_{t=1}^{T} |g(s,t) - \hat{g}(s,t)|^2, \tag{17}$$

where S and T represent pixels of an RGB image. The functions $g(s,t)$ and $\hat{g}(s,t)$ are the input and output images, respectively.

The mathematical formulation of PSNR is computed using Eq. (18).

$$\mathrm{PSNR} = 10\ .\ \log_{10} \frac{(255)^2}{\mathrm{MSE}}. \tag{18}$$

Table 2 records the correlation, MSE and PSNR values of the encrypted and decrypted images corresponding to the color image. Similarly, the correlation, MSE and PSNR values of the encrypted and decrypted images corresponding to the grayscale image are recorded in Tables 3 and 4 records the correlation, MSE and PSNR values of the encrypted and decrypted images corresponding to the binary image. The high MSE, low PSNR and low correlation values of the encrypted image validate that the encrypted image obtained corresponding to three types of image data is completely changed from the plain image data. Also, an unauthorized person cannot access the original information of the plain image from the encrypted image. On the other hand, low MSE, high PSNR and high correlation values of the decrypted image validate that the decrypted image of the proposed technique is identical to the original image and free from distortion, noise, or data loss effects.

The mathematical formulation of SSIM is computed using Eq. (19).

$$\mathrm{SSIM}(I,O) = \frac{(2\mu_I \mu_O + J_1)(2\sigma_{IO} + J_2)}{(\mu_I^2 + \mu_O^2 + J_1)(\sigma_I^2 + \sigma_O^2 + J_2)}, \tag{19}$$

Here the input and output images are denoted by I and O, respectively. The notation μ_I and μ_O denote the mean of I and O, respectively, σ_I and σ_O symbolize the standard deviation of I and O, respectively, σ_{IO} denotes the covariance between I and O, $J_1 = (k_1 L)^2$, $J_2 = (k_2 L)^2$ and $k_1 = 0.01$, $k_2 = 0.03$ and $L = 2^{\text{number of bits per pixel}} - 1$.

Table 6 records the results of the SSIM of each plane of the encrypted image. The lower numerical value of SSIM ensures that an attacker cannot get any information about the original data. Moreover, the SSIM values of each plane of the complete decrypted image with reference to the original image are listed in Table 6. The results exhibit that without any loss of information, the original image is completely recoverable.

Table 7 displays the correlation coefficients of three RGB images in the horizontal (H), vertical (V) and diagonal (D) directions. Table 8 shows the correlation coefficient of the decrypted images corresponding to three RGB images. Therefore, by analyzing all test results, it is clear that the proposed technique has strong immunity to resist statistical-based attacks.

Table 3
Correlation coefficient, MSE and PSNR of encrypted and decrypted images of grayscale image.

Parameters	Encrypted image			Decrypted image		
	R	G	B	R	G	B
Correlation	−0.0011	−6.7627e−04	−0.0023	1	1	1
MSE	4.2109e+04	3.6745e+04	4.4037e+04	0	0	0
PSNR	1.8871	2.4788	1.6926	Inf	Inf	Inf

Table 4
Correlation coefficient, MSE and PSNR of encrypted and decrypted images of binary image.

Parameters	Encrypted image			Decrypted image		
	R	G	B	R	G	B
Correlation	0.0025	−0.0016	−3.6460e−04	1	1	1
MSE	2.9123e+04	2.5608e+04	3.6427e+04	0	0	0
PSNR	3.4885	4.0471	2.5165	Inf	Inf	Inf

Table 5
Correlation, MSE and PSNR of decrypted image of RGB image for analysis-I & II.

Parameters	Analysis-I			Analysis-II		
	R	G	B	R	G	B
Correlation	0.0014	5.1821e−04	0.0022	−0.0018	0.0034	0.0099
MSE	1.0368e+04	1.0575e+04	1.0527e+04	1.0403e+04	1.0540e+04	1.0467e+04
PSNR	7.9737	7.8879	7.9078	7.9591	7.9025	7.9326

Table 6
SSIM values of encrypted and decrypted images.

Parameter	Encrypted image			Decrypted image		
	R	G	B	R	G	B
SSIM	−5.6372e−04	−6.1208e−04	−6.0504e−04	1	1	1

Table 7
Correlation analysis of original image in shift axis.

	Baboon			Lena			Peppers		
	R	G	B	R	G	B	R	G	B
H	0.9223	0.8837	0.9302	0.9753	0.9748	0.9532	0.9797	0.9903	0.9775
V	0.8596	0.7947	0.8801	0.9871	0.9872	0.9741	0.9795	0.9906	0.9785
D	0.8407	0.7631	0.8597	0.9634	0.9630	0.9334	0.9651	0.9828	0.9621

Table 8
Correlation analysis of decrypted image in shift axis.

	Baboon			Lena			Peppers		
	R	G	B	R	G	B	R	G	B
H	0.9150	0.8738	0.9231	0.9734	0.9707	0.9508	0.9737	0.9875	0.9736
V	0.8532	0.7839	0.8738	0.9848	0.9839	0.9709	0.9735	0.9880	0.9747
D	0.8349	0.7532	0.8530	0.9618	0.9583	0.9332	0.9590	0.9797	0.9595

6.3. Entropy

To estimate the uncertainty of the data, entropy analysis is used. The following equation gives the mathematical expression for entropy.

$$H(e) = -\sum_{k=1}^{M} p(e_k) \log_2 p(e_k), \tag{20}$$

Here the probability of symbol e_k is represented by $p(e_k)$. A secure cryptosystem should have resistant to entropy attacks. This is only achievable if the entropy values converge toward the ideal value of 8. Table 9 shows the entropy values of the original, encrypted and decrypted images. The test results validate that the proposed technique has the ability to resist entropy-based attacks.

Table 9
Entropy values of RGB planes of the original, encrypted and decrypted images.

Original image			Encrypted image			Decrypted image		
R	G	B	R	G	B	R	G	B
7.7827	7.7109	7.6899	5.1203	5.3577	4.9083	7.7827	7.7109	7.6899

6.4. Key space of the proposed technique

To attain the strong resistivity of the encryption scheme to brute force attacks, the key space should be large. In the proposed technique, we have used 18 secret keys. The discussion of these keys is done in three parts.

1. In the first level of encryption, AHC uses multiplicative and additive keys as secret keys. The multiplicative keys M_R, M_G, M_B are taken from $SL_m(F_p)$ domain and additive keys A_R, A_G, A_B are taken from $M_m(F_p)$ domain. The possible key space of three multiplicative keys M_R, M_G, M_B is given below.
$$3 \times \frac{\prod_{j=0}^{m-1}(p^m - p^j)}{p-1},$$
The possible key space of three additive keys A_R, A_G, A_B is given below.
$$3 \times p^{n^2},$$
Therefore, the total number of possible keys of AHC is
$$9 \times p^{n^2} \times \frac{\prod_{j=0}^{m-1}(p^m - p^j)}{p-1}.$$
The larger value of m and p it gets, the more enormous the key space.
2. In the second level of encryption, RP2DFrHT uses six fractional orders with the precision of 10^{-4} therefore the size of key space is $10^{4\times 6} = 10^{24}$.
3. In the last level of encryption, AM uses six Arnold secret keys which can take infinitely many values

Therefore, our proposed technique has an exorbitant huge key space which is large enough to make brute force attacks unfeasible.

6.5. Robustness to known-plaintext attack (KPA) and chosen-ciphertext attack (CCA)

This section discusses the robustness of the proposed technique to KPA and CCA. In KPA, the original and encrypted images are accessible to the cryptanalyst and he/she tries to find confidential information from this information. On the other hand, in CCA, the cryptanalyst knows about the encryption mechanism and he/she tries to generate an encrypted image corresponding to the randomly chosen original image. The proposed technique has the ability to resist both KPA and CCA because the security of the proposed technique depends not only upon the secret keys but also upon the correct arrangements and positions of used parameters. Also, the proposed technique has exorbitant huge key space and high key sensitivity towards its encryption and decryption keys.

7. Comparison analysis

This section gives a comparison between the proposed technique and similar existing techniques [9,10,14–19,23–25].

S.No.	Existing techniques [9,10,14–19,23–25]	Our proposed technique
1.	The existing techniques can encrypt single or double images only.	Our proposed technique can encrypt multiple images simultaneously.
2.	Although the existing techniques are multiple image encryption, yet, merely applicable either for RGB or grayscale images.	The proposed technique is applicable for three types of image data such as RGB, grayscale and binary images.
3.	The existing techniques increase the dimension of the encrypted data.	The dimension of the encrypted data of the proposed technique is equal to the original data.
4.	To secure the existing techniques, only secret keys are utilized.	To secure the proposed technique, secret keys, arrangements as well as positions of used parameters are utilized.
5.	The existing techniques possess single layer security.	The proposed technique possesses multi-layer security.

8. Conclusion

In this paper, we have proposed a MIE technique based on AHC, RP2DFrHT and 2D AM. This technique facilitates the single-channel processing for each image by converting the three color images into three indexed images. The indexed images are taken as R, G and B planes of the single color image. In the first level of encryption, AHC is applied on the RGB planes and provides strong degree of confusion and diffusion operations. Then, RP2DFrHT is applied on the partially encrypted images and provides real-valued encrypted images. This process also improves the efficiency of the proposed technique. In the last stage, the resultant image pixels are scrambled using 2D AM. This enhances the randomness of the pixels and enlarges the key space of the system. Our technique is tested on three types of multiple images such as color, grayscale and binary images. Earlier developed techniques can secure single image either in the time domain, frequency domain, or co-ordinate domain, but the proposed technique can secure multiple images in the time, frequency and co-ordinate domains. Earlier developed techniques provide only one layer of security with the secret keys, but the proposed technique provides multi-layer security with the secret keys, arrangements and positions of used parameters. The results of simulation analysis, security analysis and statistical analysis validate that the proposed technique has less space complexity, high security, high efficiency and low time complexity and has the capability to resist statistical-based attacks, brute force attacks and cryptographic-based attacks.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to https://doi.org/10.1016/j.compeleceng.2023.108609.

Data availability

No data was used for the research described in the article.

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Shazia Sabir is currently pursuing the Ph.D. degree in Mathematics from Birla Institute of Technology Mesra, Ranchi, India. Her research interests include image processing and image security.

Vandana Guleria is working as an Assistant Professor in the Department of Mathematics, Birla Institute of Technology Mesra, Ranchi, India. Her research interests include image processing and image security.